

Object Response – G-Clamp

Cost Engineering, Hostile Environments & Everyday Carry Redesigns

A comprehensive industrial design investigation deconstructing an everyday £2 50mm cast-iron G-clamp. After analysing its anatomy, manufacturing processes, ergonomics, and Bill of Materials, I re-engineered the clamp across half-price and double-price constraints, extreme environments, and portable carry.

Discipline

Industrial Design

Context

CSM Unit 1 Brief

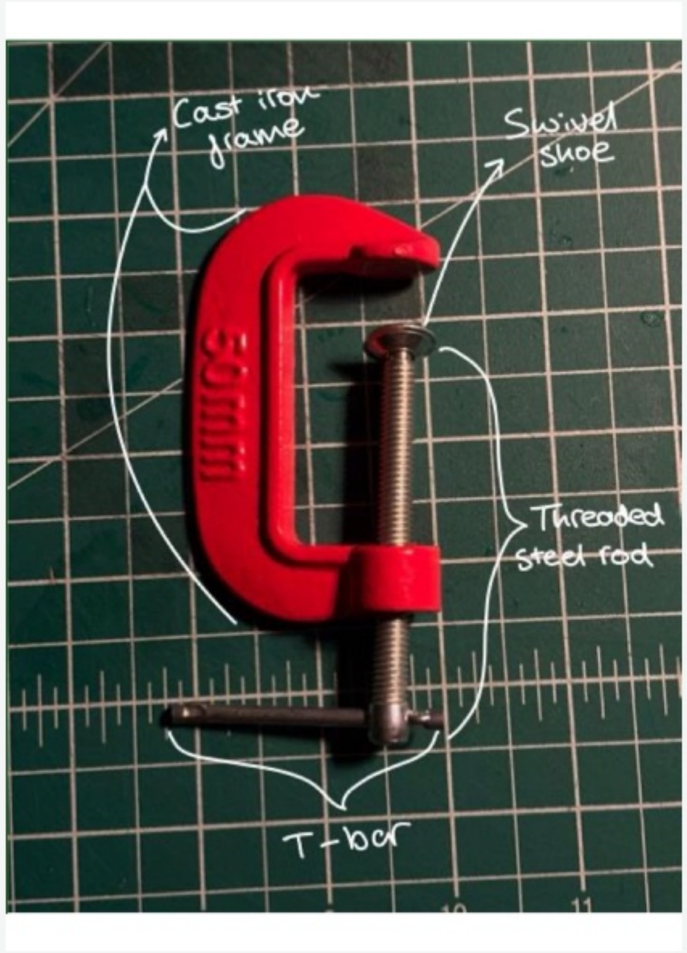
Tools

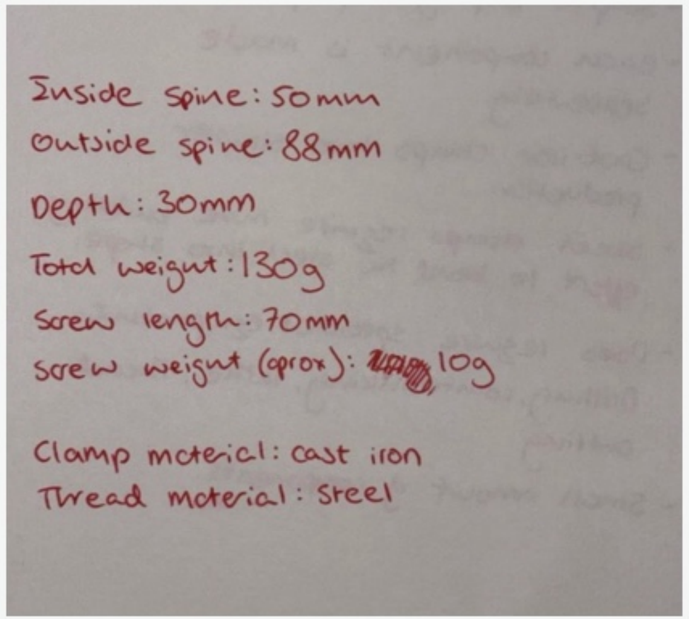
Sketching, Cardboard CAD

Outcome

Over 20 distinct mechanical redesigns validated through 1:1 functional cardboard prototypes—ranging from NASA Pistol Grip Tool (PGT) EVA sockets to carabiner and cap-brim micro-clamps.

Task 1 — product anatomy & component weights





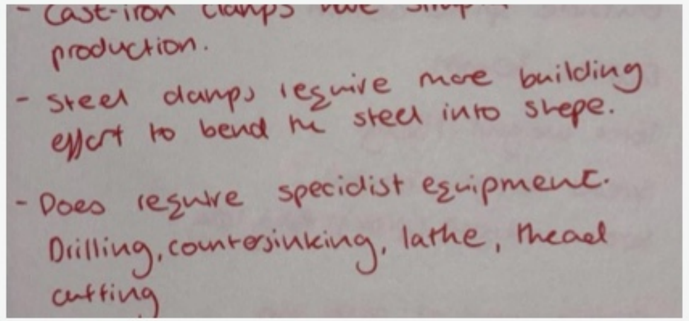


Fig. 01–03 — Annotated 50mm G-clamp anatomy, studio profile, and digital scale mass verification (130g total)

Physical Deconstruction — 50mm Cast G-Clamp

Weighing 130 grams in total, the standard 50mm workshop G-clamp consists of four metallic components assembled without removable fasteners—relying on a cold-upset tommy-bar cap and a crimped ball-and-socket swivel pad.

01 **G-Frame Spine (97 g)** — Sand-cast ductile/grey iron frame with I-beam web cross-section and dipped red enamel paint.

02 **Threaded Screw Rod (23 g)** — 70mm long M8 acme/metric rolled steel thread with drilled through-hole head.

03 **Swivel Shoe Pad (5 g)** — Stamped low-carbon steel cup crimped over the ball-end of the threaded rod.

04 **Tommy Bar / T-Bar (5 g)** — 55mm mild steel sliding lever rod with upset ends.

Task 1 — purpose, performance & ergonomics

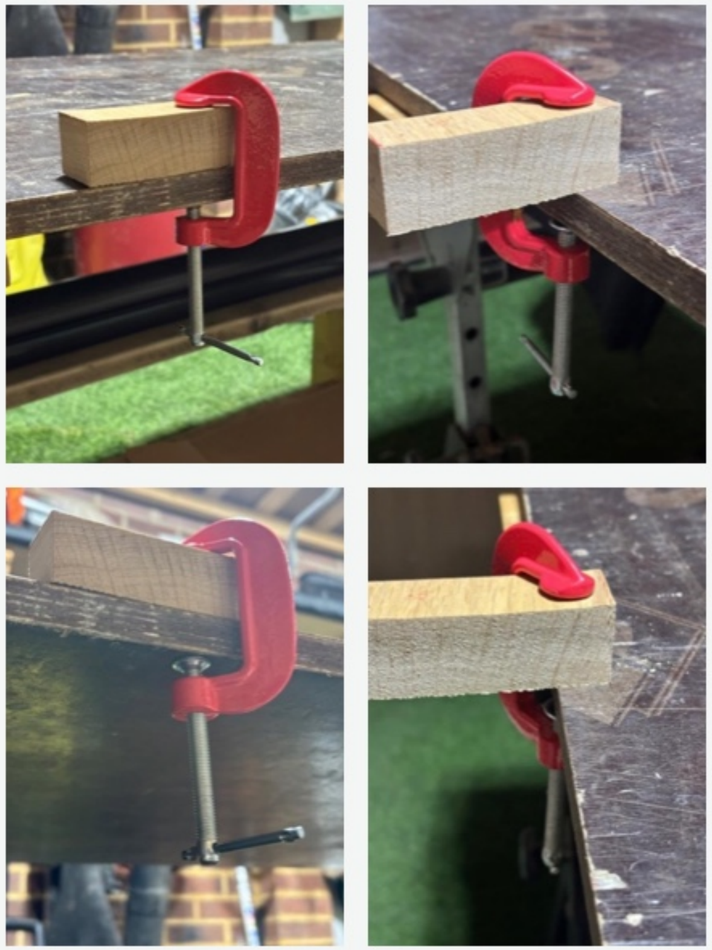


Fig. 04–07 — Clamping force tests, red paint transfer defect on timber, and two-handed ergonomic grip friction

01 — Purpose

Designed to hold wood, metal, or plastic workpieces firmly together or to a workbench during gluing, sawing, drilling, and welding.

02 — Performance

Delivers high static clamping force for a £2 tool, but the brittle red paint flakes off onto clean timber under pressure, and the un-machined cast anvil face mars softwoods.

03 — Ergonomics & Usability

Awkward two-handed operation: one hand must hold the heavy frame while the other spins the gritty thread. The thin 4mm steel T-bar digs painfully into fingers at high torque.

Task 1 — Bill of Materials, appearance & sustainability

Part	Component	Mass	Material	Manufacturing Process	Function & Surface Finish
01	**G-Frame Spine**	97 g	Cast / Ductile Iron	Green-sand casting + fettling	Resists bending moments; dipped red gloss paint protects against rust
02	**Threaded Rod**	23 g	Medium Carbon Steel	Cold heading + thread rolling	Converts rotary torque into linear clamping force; black oxide finish
03	**Swivel Jaw Pad**	5 g	Low-Carbon Sheet Steel	Blanking, deep drawing & crimping	Self-aligns on angled surfaces without spinning on the workpiece
04	**Tommy Bar**	5 g	Mild Steel Wire Rod	Wire drawing, cutting & upsetting	Provides sliding lever arm for tightening in confined corners

04 — Aesthetic & Industrial Appearance

Utilitarian workshop archetype: the bright pillar-box red enamel contrasts against the blackened steel screw, visually signalling a heavy-duty tool that is easy to spot among wood shavings. However, rough sand-cast parting lines and paint drips betray its ultra-low-cost mass production.

05 — Sustainability & End-of-Life

High intrinsic longevity—a cast-iron frame can last decades—yet the Permanent Crimped Assembly makes non-destructive disassembly impossible. If the mild-steel screw bends or strips, the entire 130g clamp is scrapped, though 100% of its ferrous mass can be melted down in electric-arc steel recycling.

Task 2 — half-price & double-price breadth ideation

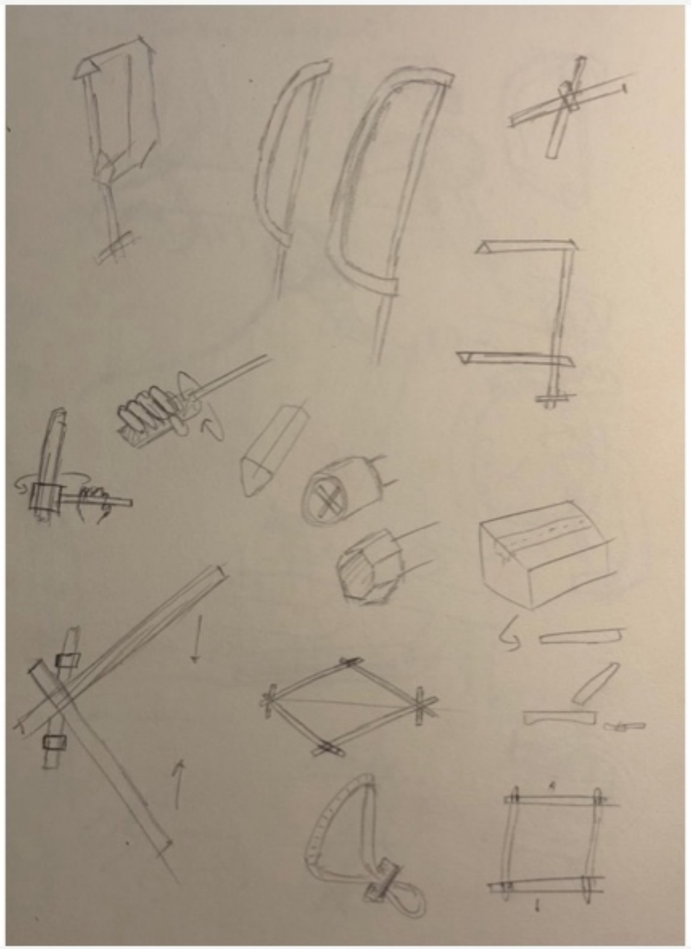


Fig. 08 — Half-Price (£1) ideation sheet: stamped sheet, folded wire, polymer grips

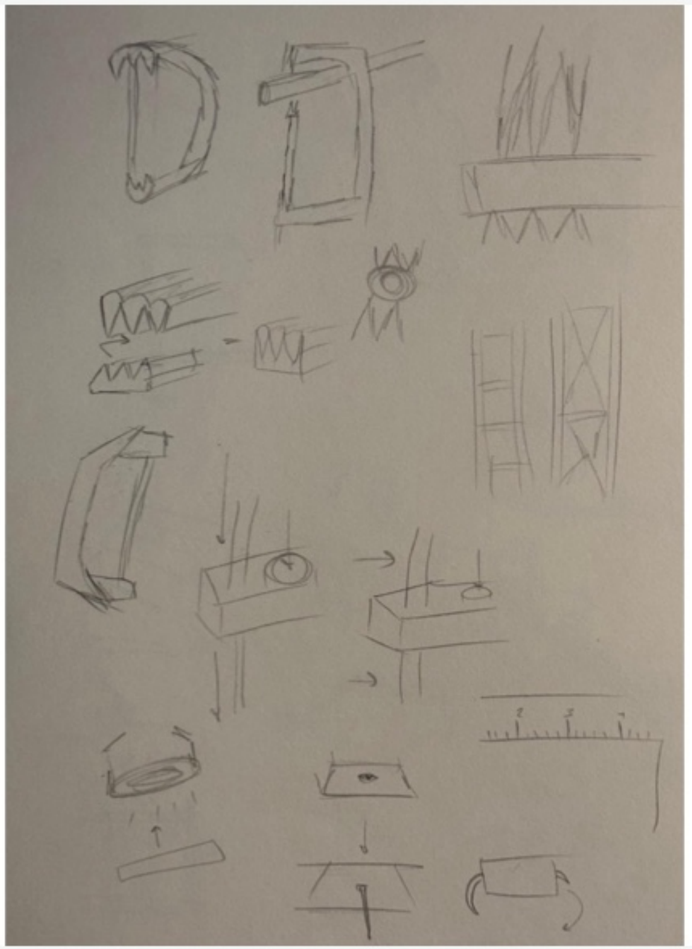
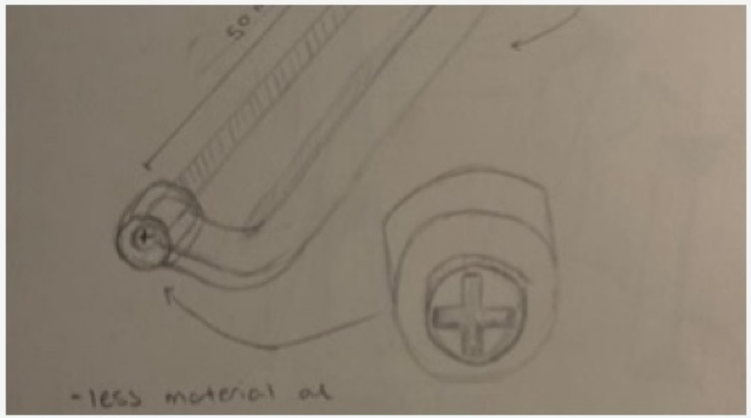


Fig. 09 — Double-Price (£4) ideation sheet: pipe jaws, magnets, scissor linkages

Task 2.1 — half-price redesign sketches



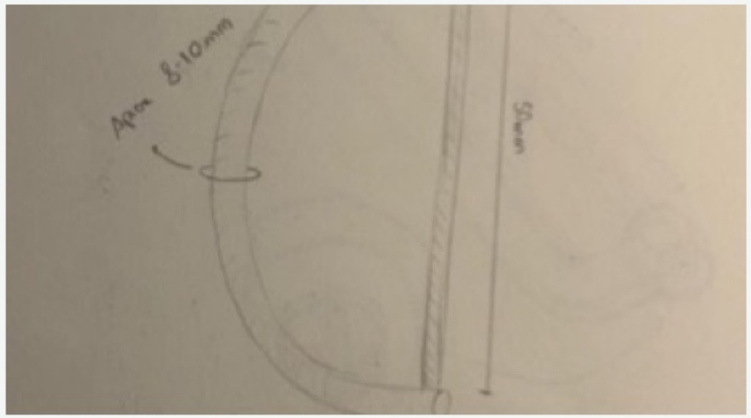
01 — Stamped Sheet-Steel Clamp



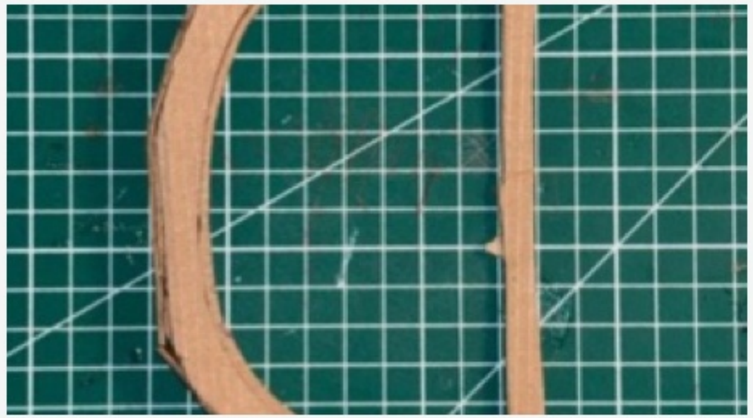
02 — Folded Steel Rod Clamp



03 — Moulded Polymer T-Grip



04 — Laminated Strip Spine

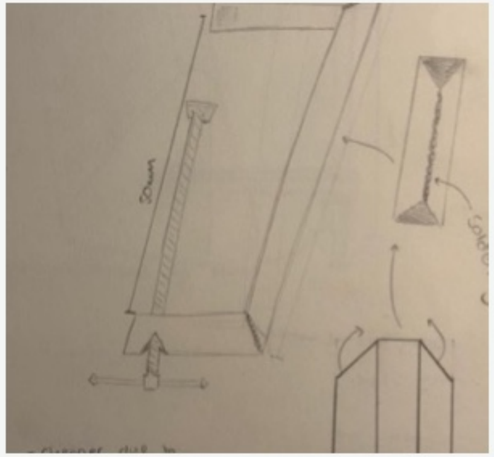


05 — Wire-Formed Spring Assist

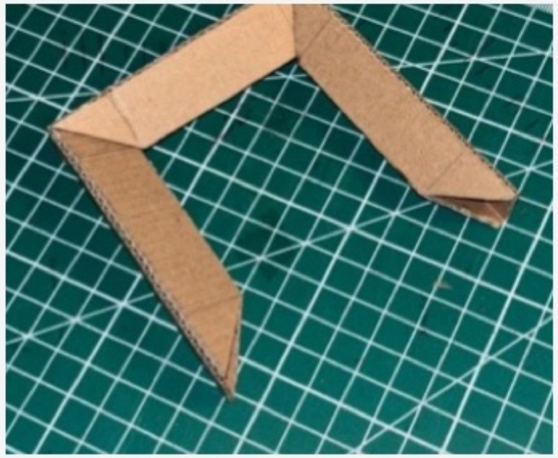


06 — Hex-Head Spanner Drive

Task 2.1 — half-price models: sheet cut & folded rod

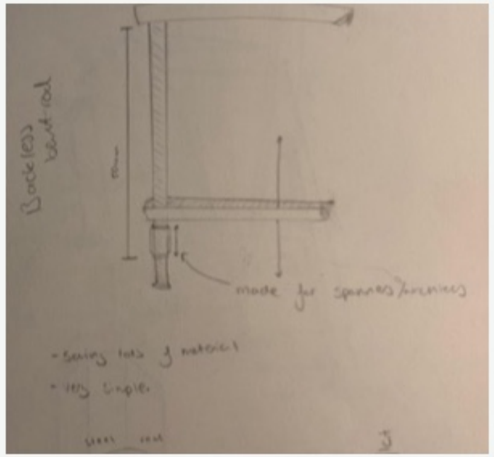




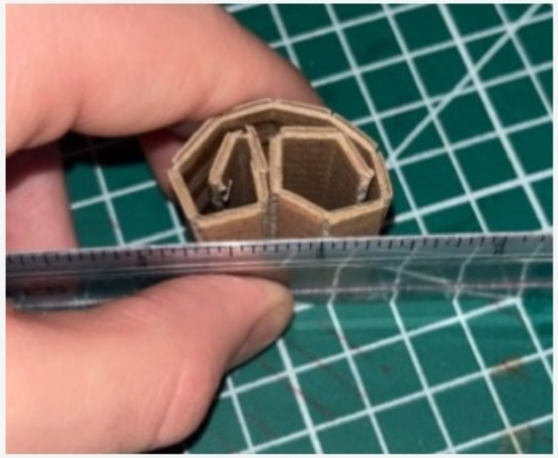


01 — Sheet Cut Clamp

Stamped from flat sheet steel and folded into rigid triangular prism beams—replacing slow sand-casting with high-speed progressive die stamping.



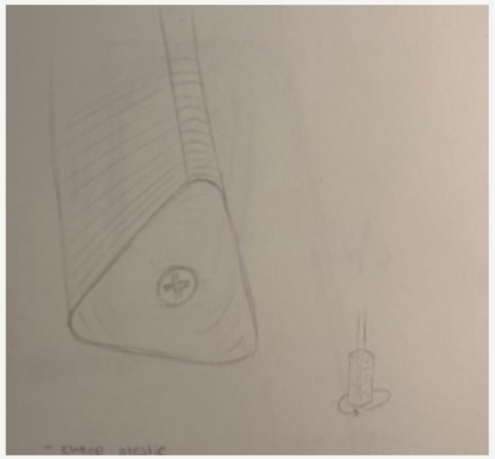


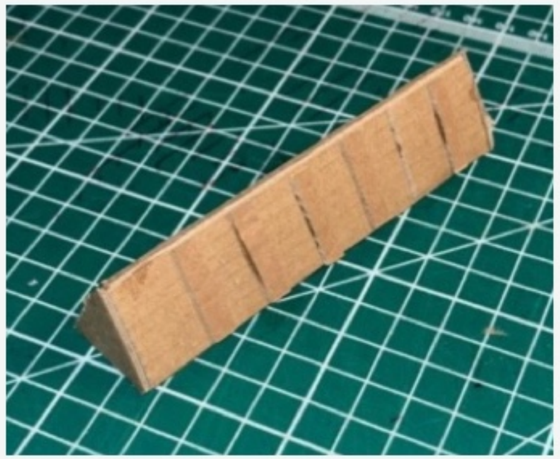


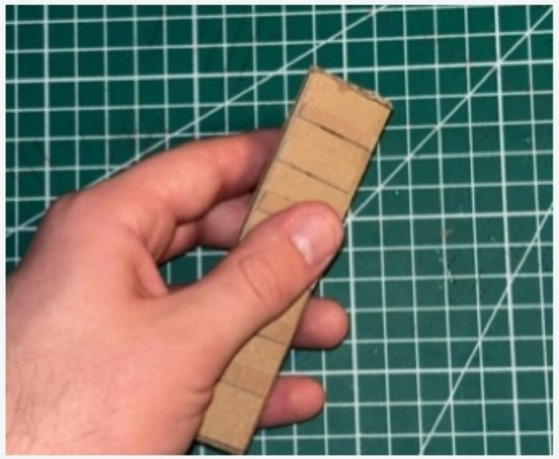
02 — Folded Steel Rod

Formed from bent steel rod and tightened with a standard hex spanner—eliminating the cast spine, swivel cup, and sliding T-bar assembly.

Task 2.1 — half-price models: polymer grip & laminated strips

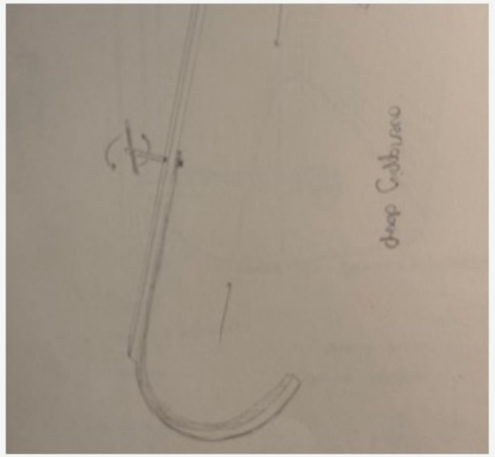






03 — Polymer Grip

Replaces the cross-drilled steel T-bar with an over-moulded triangular polymer knob—cheaper to manufacture, rust-proof, and kinder on fingers.



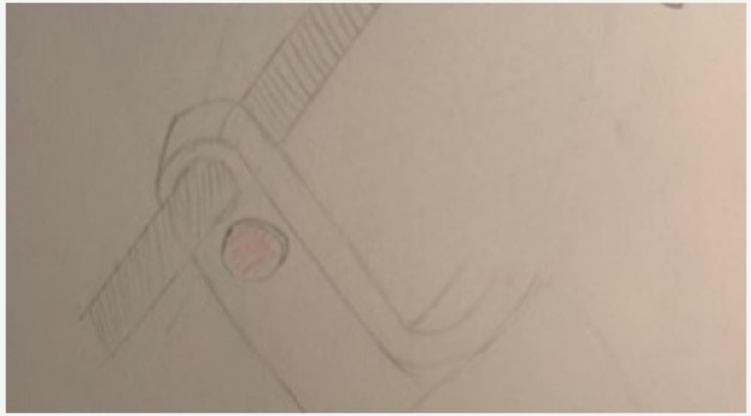




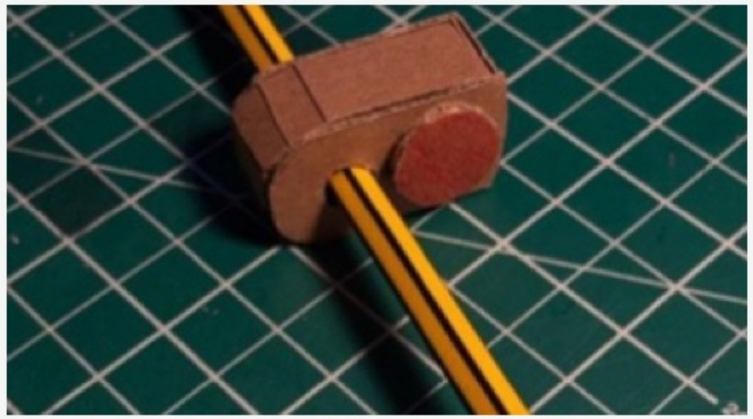
04 — Laminated Strips

Builds the deep-throat frame by riveting stamped flat steel laminations together, eliminating foundry tooling costs.

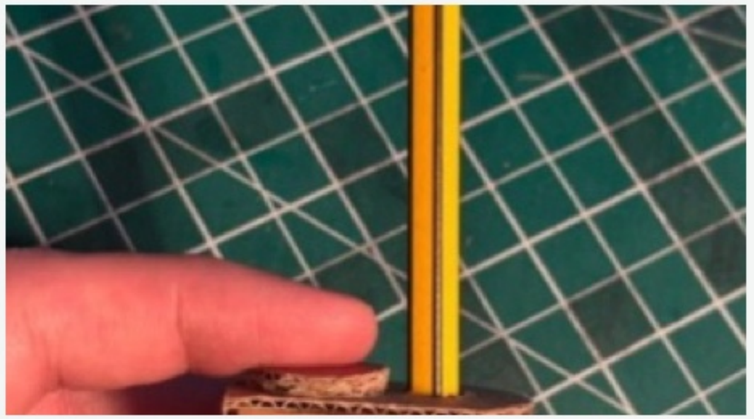
Task 2.2 — double-price redesign sketches



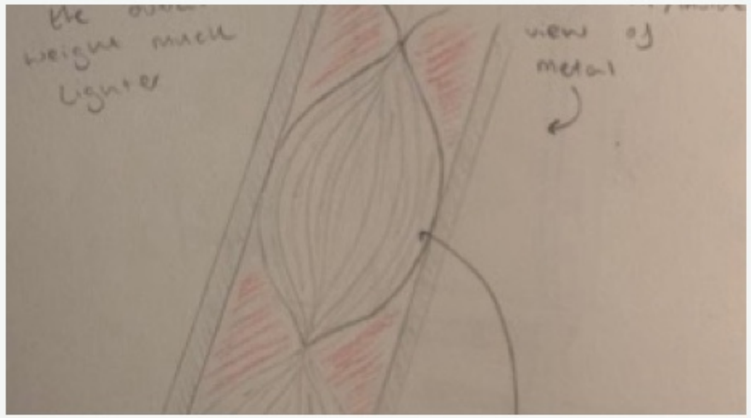
01 — Serrated Pipe-Grip V-Jaw



02 — Recessed Neodymium Magnet Jaw



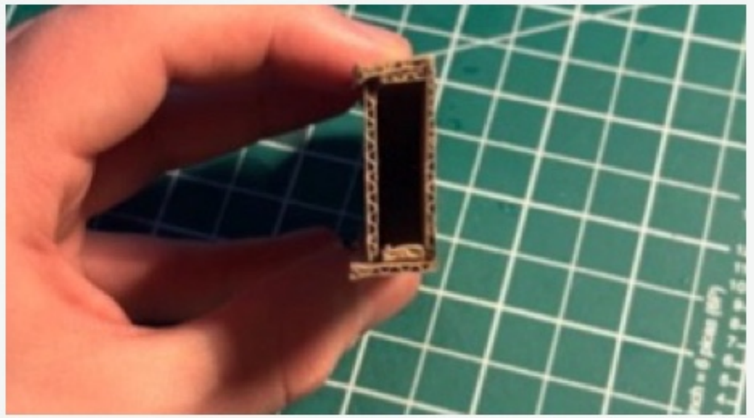
03 — Scissor-Lift Pantograph Linkage



04 — Deep-Throat Knurled T-Bar

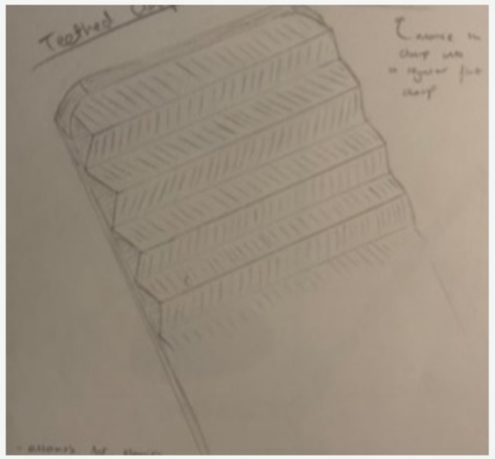


05 — Quick-Release Ratchet Trigger



06 — Dual-Axis Corner Alignment Clamp

Task 2.2 — double-price models: pipe jaw & magnetised jaw

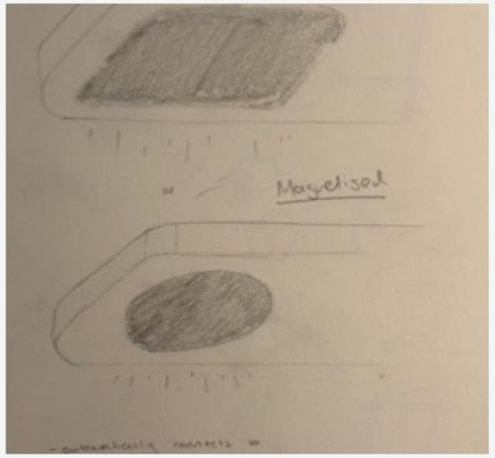






01 — Teethed Pipe Jaw

Integrates a serrated V-channel into the fixed jaw to grip cylindrical pipes and dowels without slipping, paired with a snap-on flat pad for timber.



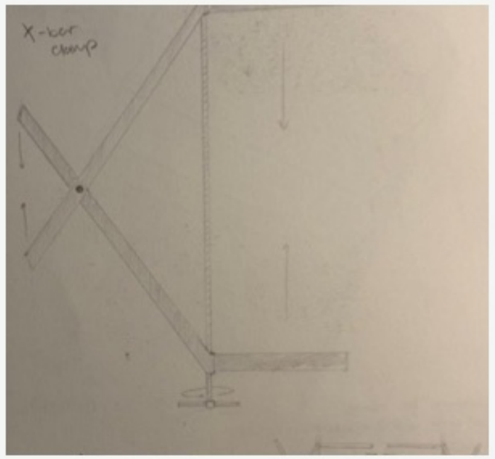


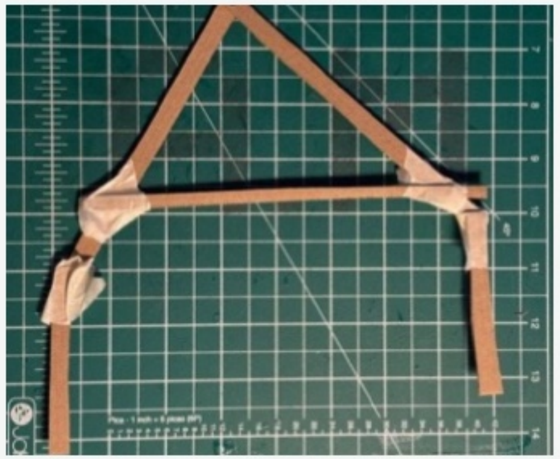


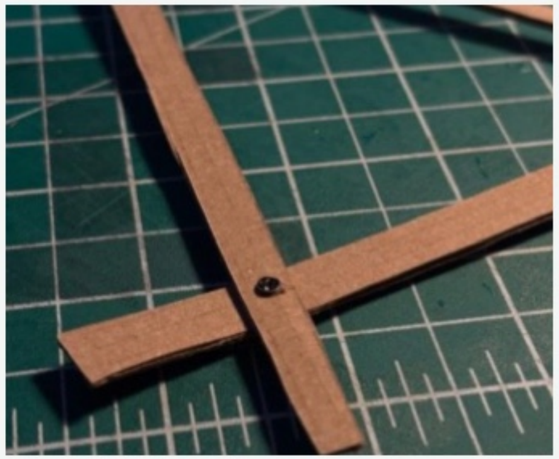
02 — Magnetised Jaw

Embeds a flush neodymium magnet in the fixed anvil face so welders can position the clamp one-handed against steel stock.

Task 2.2 — double-price models: scissor lift & deep throat

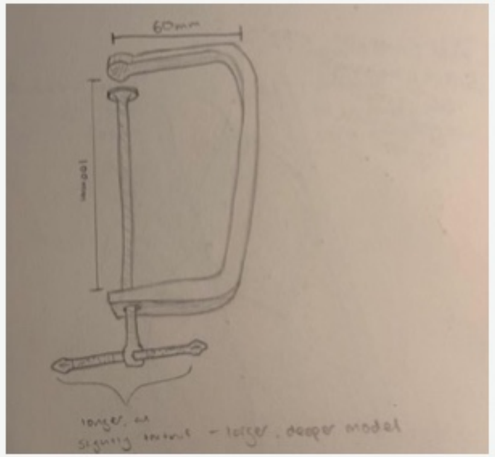


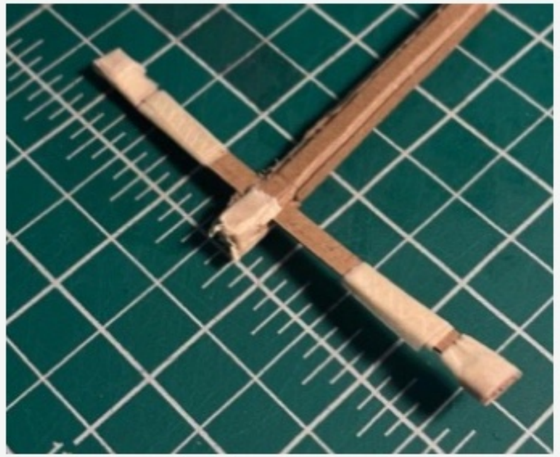




03 — Scissor-Lift Clamp

Replaces the exposed acme screw with a pantograph scissor linkage—keeping the drive screw clear of glue drips while delivering parallel jaw travel.







04 — Deep-Throat & Knurled Bar

Extends throat reach to 100mm for clamping wide panels and adds thickened, knurled T-bar ends for comfortable high-torque tightening.

Task 3.1 research — hostile environment: underwater



Fig. 34 — Commercial saturation diver working in thick neoprene gloves with tethered pneumatic/hydraulic rigging

Underwater Engineering Constraints

Commercial divers performing subsea pipeline, hull, and dock repairs operate in cold, low-visibility water wearing 5–7mm neoprene gloves or drysuit mitts that eliminate fine finger dexterity.

01 **Gloved Dexterity Loss** — A tiny 4mm sliding T-bar is impossible to pinch or spin inside thick diving gloves.

02 **Negative Buoyancy Drop Hazard** — In open water, an untethered 130g iron tool sinks immediately into silt; every tool must clip to a BCD lanyard.

03 **Saltwater Galvanic Corrosion** — Standard painted cast iron rusts and seizes rapidly in marine environments.

Task 3.1 research — hostile environment: outer space EVA



Fig. 35 — NASA Pistol Grip Tool (PGT) schematic and astronaut EVA Mini Workstation (MWS) chest tether system

Microgravity & Pressurised EVA Suit Constraints

During Extra-Vehicular Activity (EVA), astronauts work in rigid 4.3 psi pressurised gloves where repetitive wrist rotation causes rapid forearm exhaustion, and Newton's third law causes unbraced torque to spin the astronaut.

01 **NASA PGT Standard Interface** — EVA tools use a standardised 3/8-inch hex drive socket turned by the computerised Pistol Grip Tool.

02 **MWS Velcro & Tether Retention** — Tools must dock onto the astronaut's chest Mini Workstation or Velcro wrist pad between operations.

03 **Vacuum Cold-Welding** — Uncoated metals gall and cold-weld in hard vacuum; dry lubricants and anodised interfaces are mandatory.

Task 3.1 research — hostile environment: underground



Fig. 36 — Underground mining and tunnelling maintenance in zero-ambient-light, high-particulate conditions

Subterranean Tunnelling & Mining Constraints

Underground maintenance engineers work in 0 lux ambient darkness illuminated only by narrow cap-lamp beams, surrounded by abrasive rock dust, mud slurry, and airborne grit.

01 **Thread Seizure from Grit** — High Particulate Matter (HPM) packs into standard helical screw threads, jamming the clamp solid.

02 **Zero-Light Tool Location** — Dropped or clamped dark-metal tools disappear against wet rock walls once a worker turns their head.

03 **Self-Cleaning Geometry** — Threads and jaws require axial escape channels that scrape out mud and grit on every rotation.

Task 3.3 research — portable attachment & everyday carry







Fig. 37 — Nite Ize S-Biner dual spring gate

Fig. 38 — Mirafit Aluminium Carabiner

Fig. 39 — SOG Sync II Belt-Buckle Multi-Tool

Everyday Carry (EDC) & Wearable Utility

Why does the G-clamp remain trapped on the workshop bench while pliers and knives became everyday carry tools?

Studying dual-gate S-Biners, climbing carabiners, and wearable belt-buckle tools showed that merging a spring-loaded clip gate or wearable clasp directly into the G-frame turns a heavy workshop clamp into an instant pocket utility.

Tasks 3.1 & 3.3 — hostile environment & portability breadth sheets

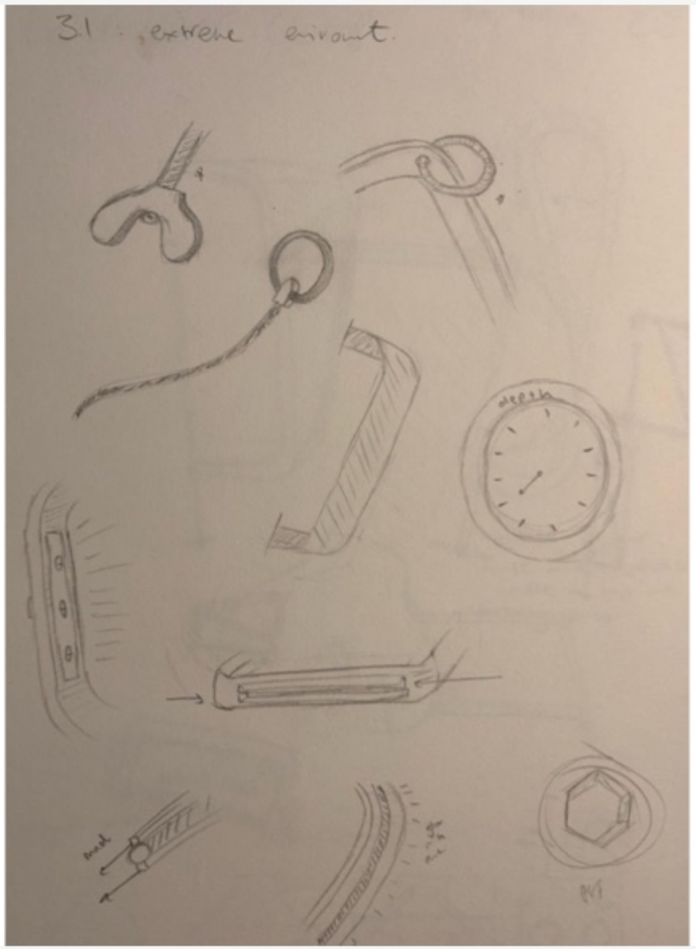


Fig. 40 — Task 3.1 Hostile Environments breadth sheet

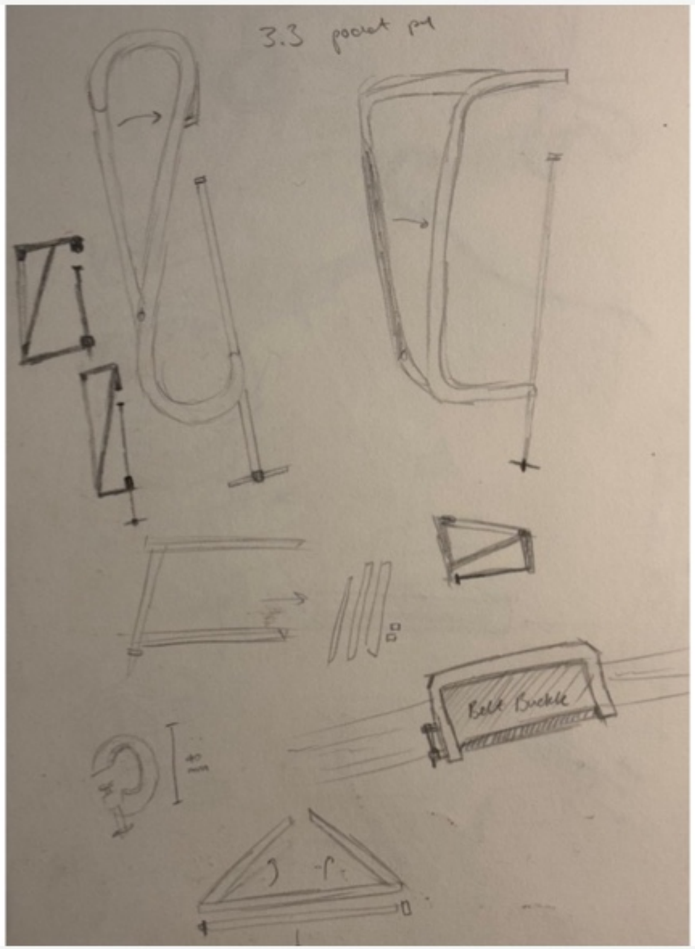
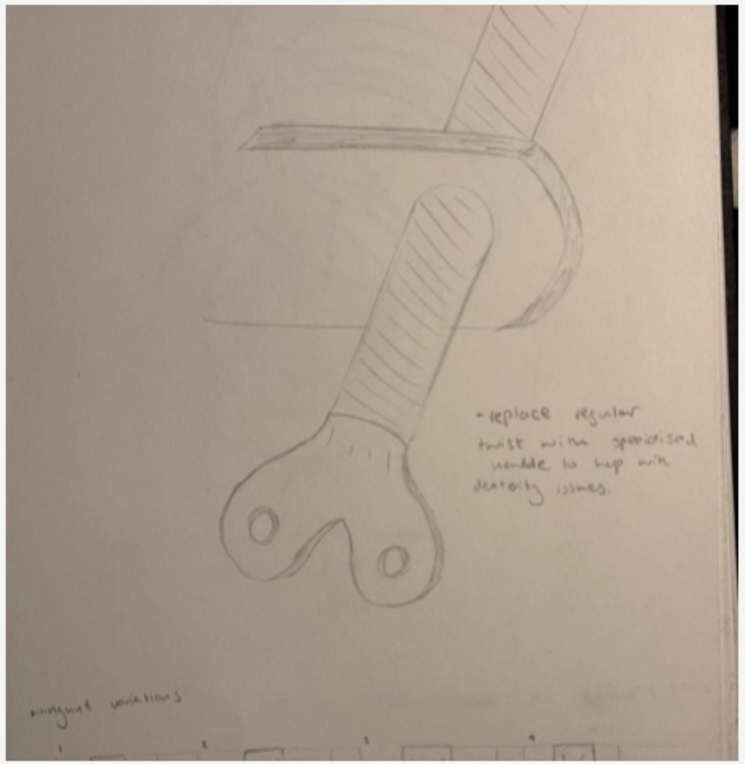


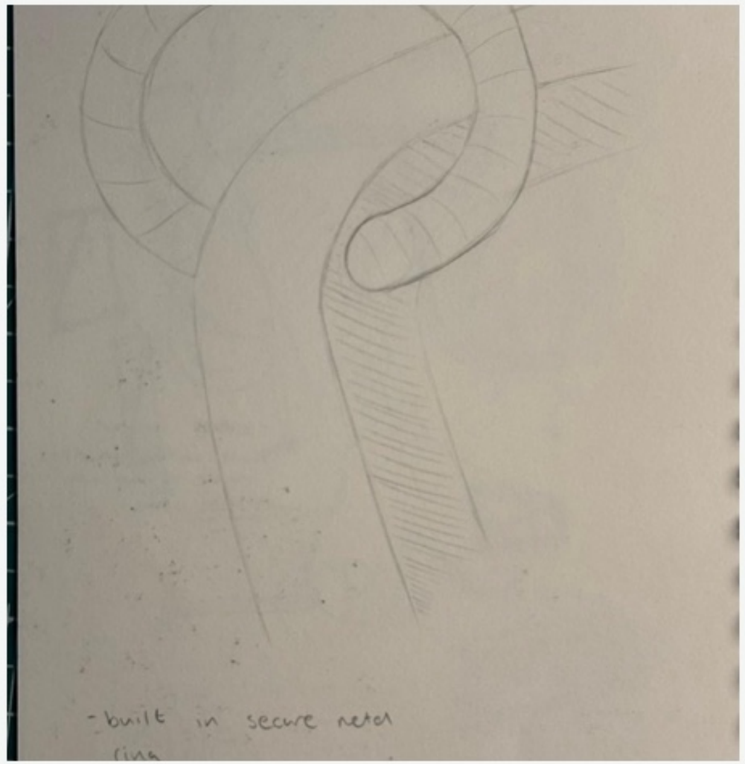
Fig. 41 — Task 3.3 Everyday Carry & Portability breadth sheet

Task 3.1 sketches — underwater & EVA adaptations



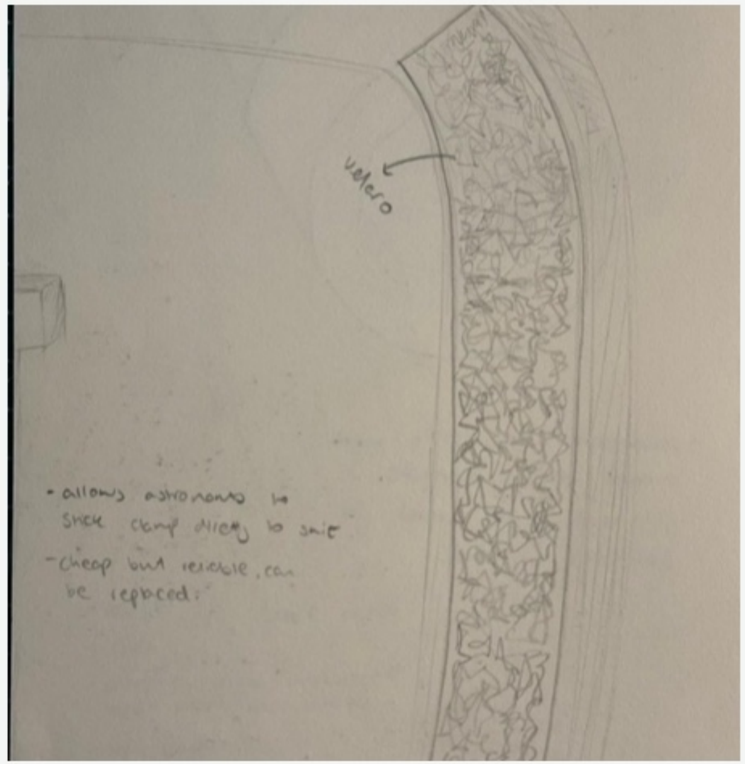
01 — Butterfly Bolt (Underwater)

Oversized winged butterfly head replaces the tiny T-bar so a diver in 7mm neoprene gloves can torque the screw using palm pressure.



02 — Built-in Tether Ring (Underwater)

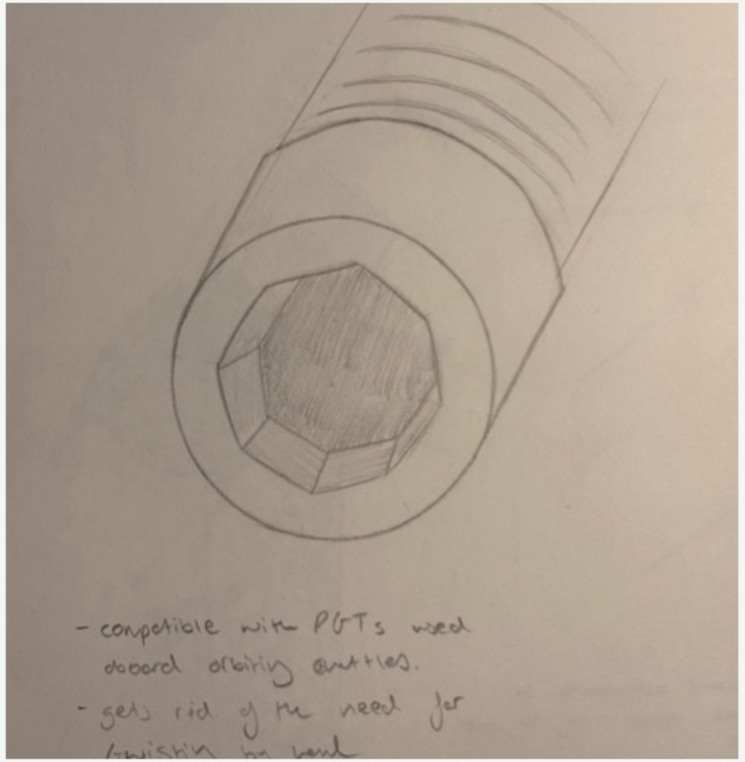
Cast dorsal eye-ring welded to the spine balances the clamp's centre of gravity when clipped to a diving carabiner.



03 — Astronaut MWS Velcro Spine (Space)

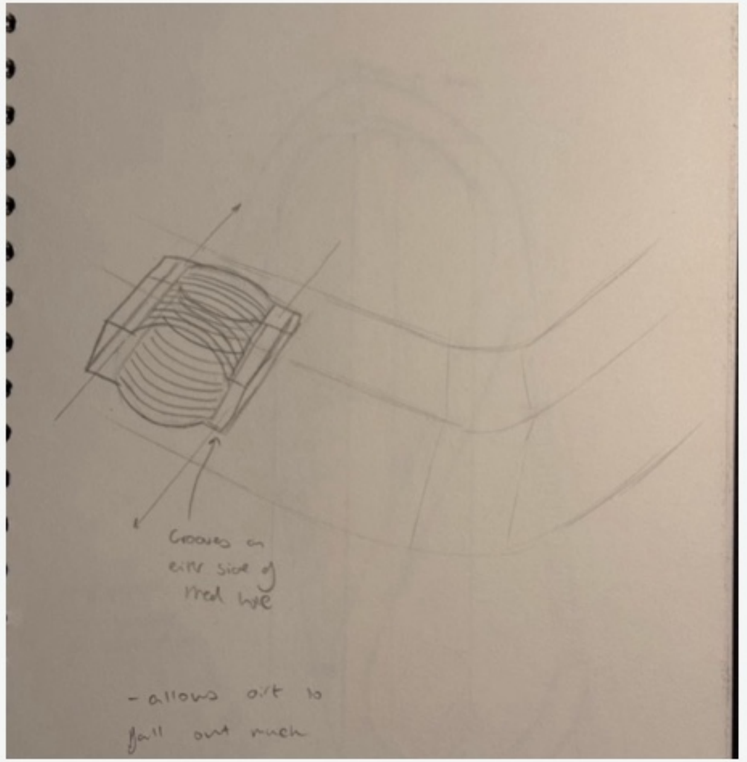
Hook-and-loop spine pad allows the clamp to dock directly onto an astronaut's EVA suit wrist or chest workstation.

Task 3.1 sketches — PGT drive & underground adaptations



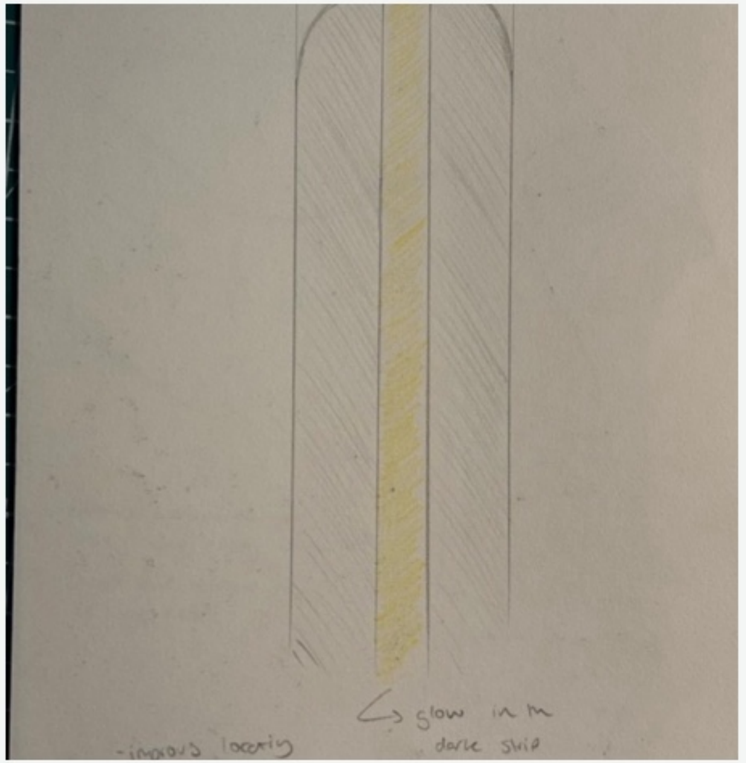
04 — NASA PGT Hex Socket (Space)

Replaces the manual T-bar with a 3/8-inch hex socket driven directly by NASA's computerised Pistol Grip Tool.



05 — Grit-Proof Slotted Thread (Underground)

Longitudinal flute cut along the screw rod scrapes mud and rock grit out of the internal nut threads on every turn.



06 — Photoluminescent Spine (Underground)

Recessed glow-in-the-dark strip along the I-beam web keeps clamped workpieces visible in 0-lux tunnel environments.

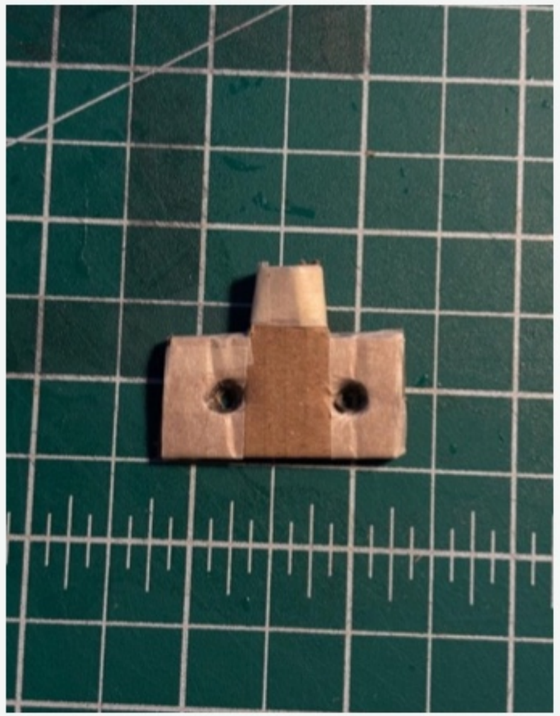
Task 3.1 models — underwater butterfly bolt (Mk1 & Mk2)

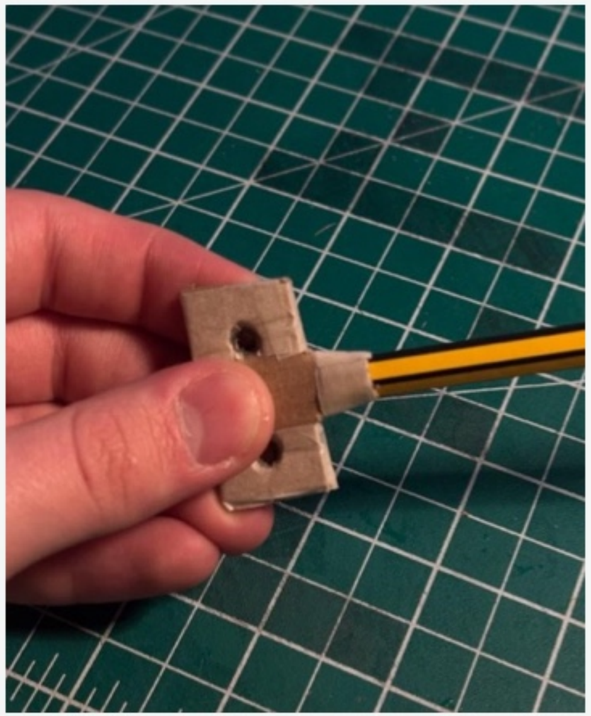




Butterfly Bolt Mk1 — Flat Wing Profile

First 1:1 cardboard prototype tested wing span against knuckle clearance; flat wings flexed under high thumb pressure.



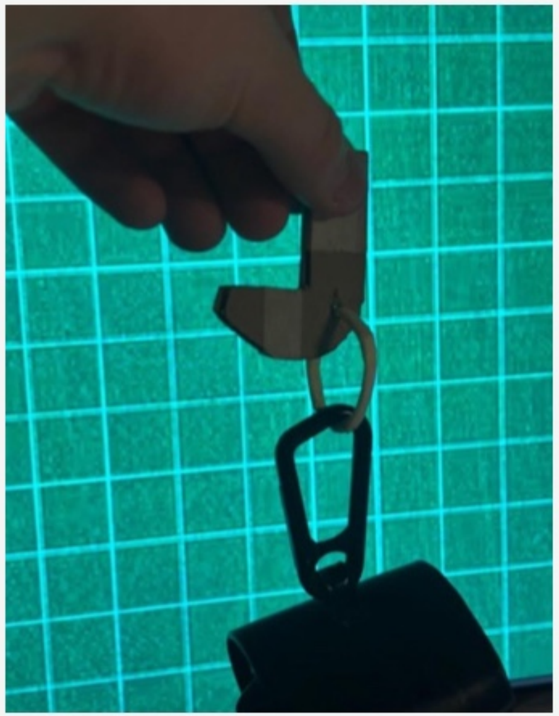


Butterfly Bolt Mk2 — Gusseted Ergonomic Wings

Second iteration added central triangular gussets and contoured finger scallops, allowing full-palm rotation with zero finger pinching.

Task 3.1 models — dive tether ring & astronaut Velcro addon





Built-in Steel Tether Ring (Underwater)

Integrated dorsal loop tested with a metal carabiner to verify one-handed clipping and balanced hanging angle on a dive harness.





Astronaut Velcro Spine Addon (Outer Space)

Wraparound hook-and-loop spine saddle tested on a sleeve to simulate zero-G retention on an astronaut's forearm.

Task 3.1 models — NASA PGT socket & grit-proof thread





NASA PGT Hex Drive Socket (Outer Space)

1:1 cardboard socket head and handheld power-driver mockup verifying coaxial engagement without side-loading the clamp.

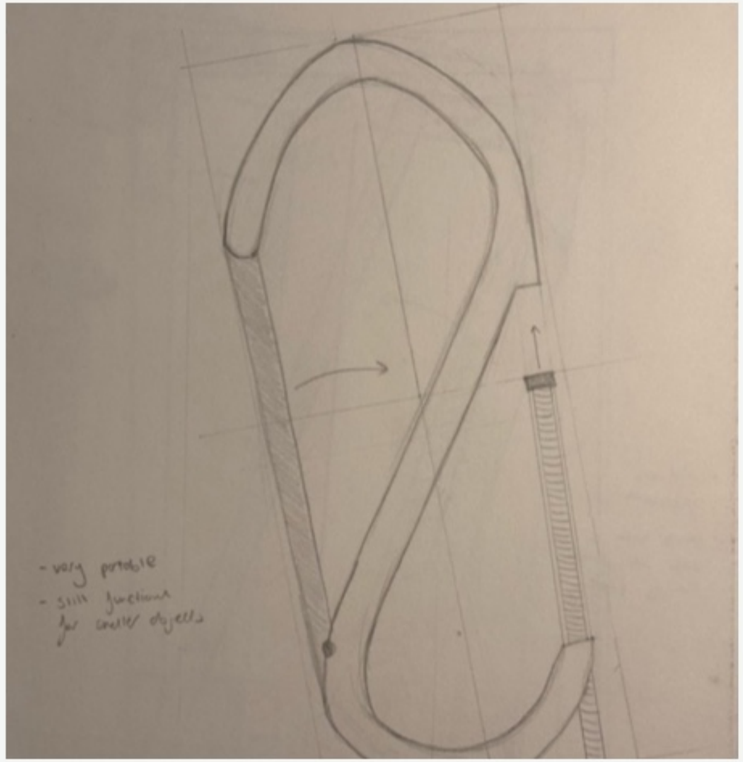




Grit-Proof Self-Cleaning Thread (Underground)

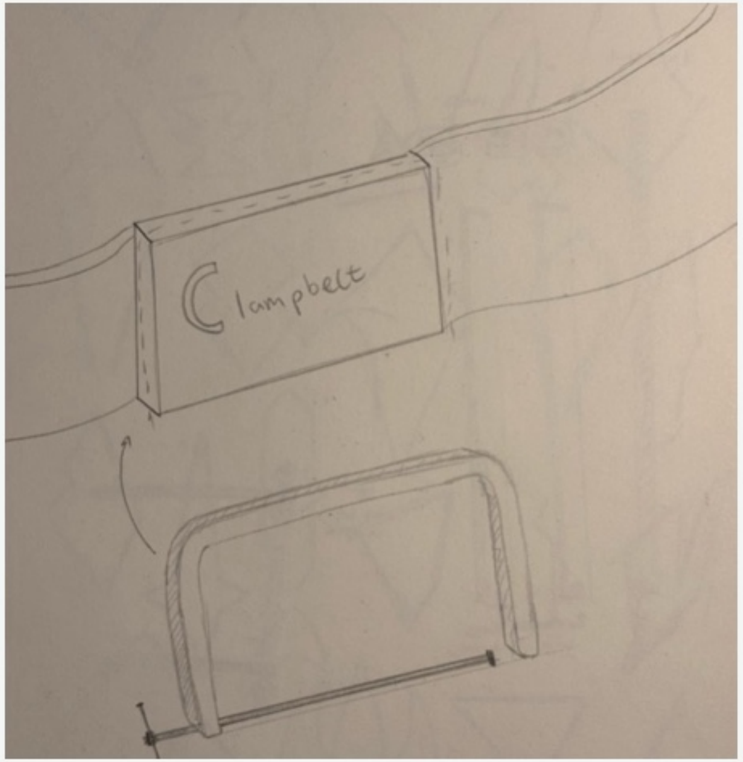
Large-scale thread section model demonstrating how the longitudinal clearing flute expels trapped dirt and rock dust.

Task 3.3 sketches — S-Biner, belt buckle & carabiner Mk1



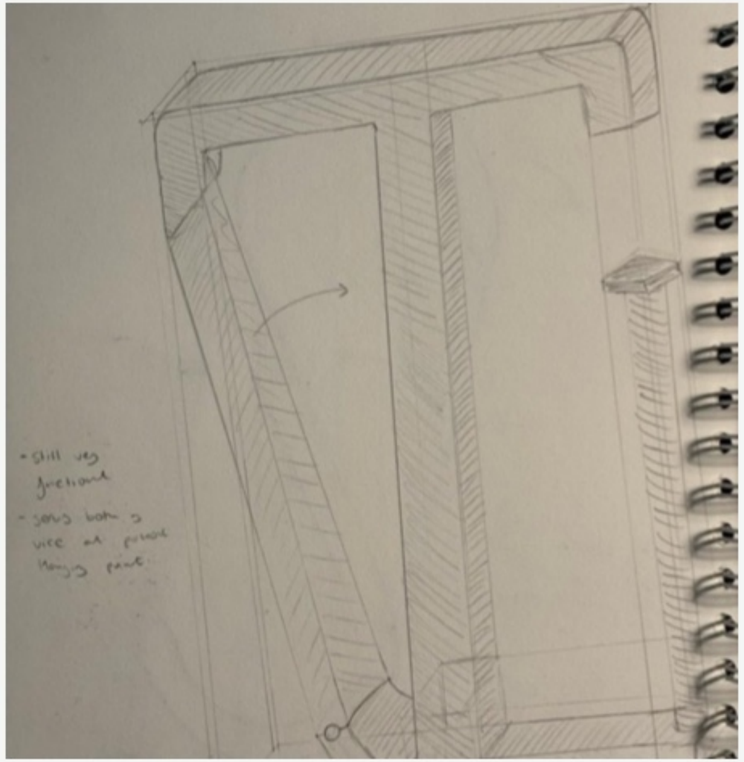
01 — S-Biner Merge Clamp

Adds a spring-loaded wire gate to the rear spine so the clamp clips directly onto belt loops or rucksack webbing.



02 — Belt-Buckle Clamp

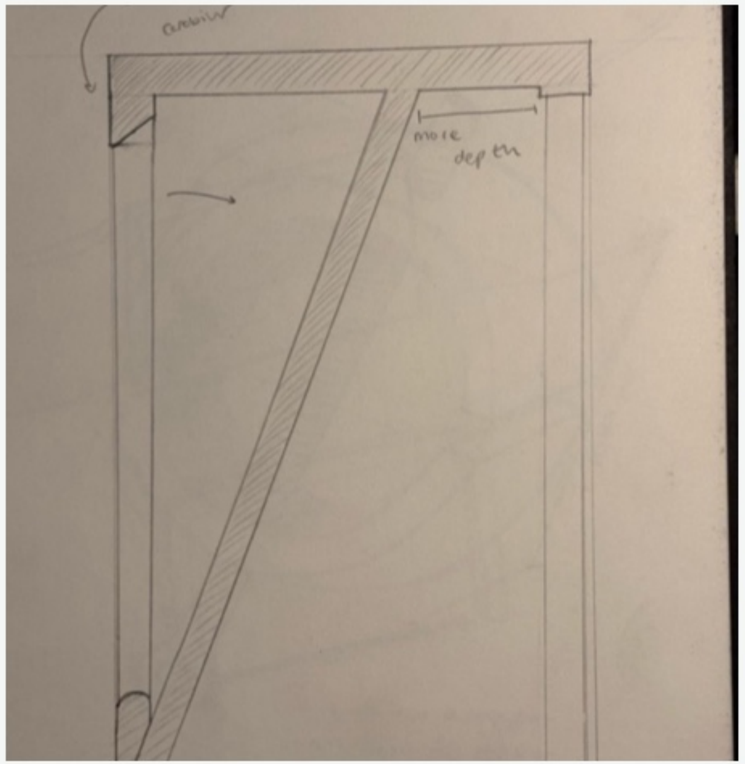
Re-proportions the G-frame into a flat rectangular frame that functions as an everyday trouser belt buckle until needed.



03 — Carabiner Clamp Mk1

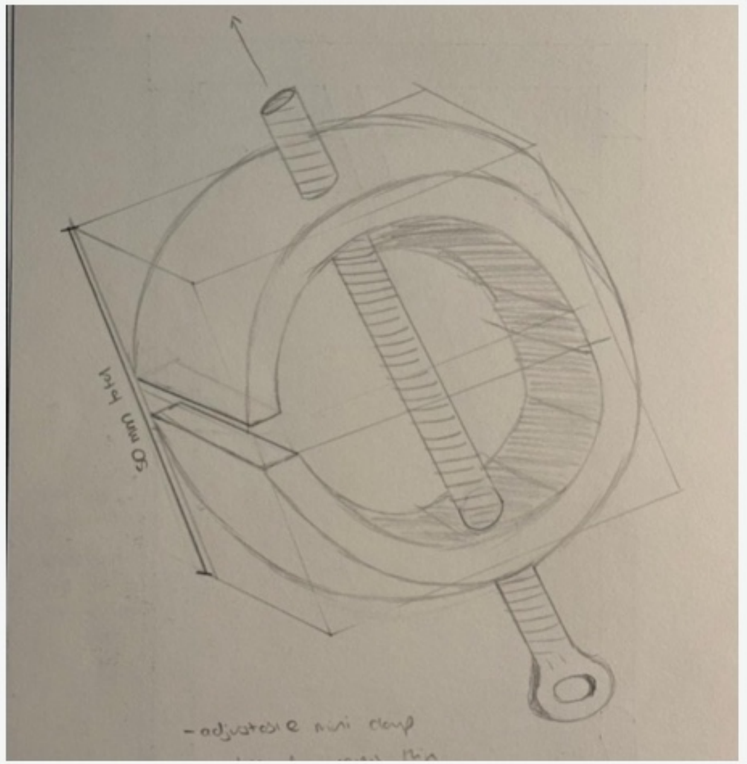
Integrates the clamping screw parallel to the carabiner spine, turning the upper hook into a dual-purpose hanging and clamping jaw.

Task 3.3 sketches — carabiner Mk2, mini clip & flat-pack



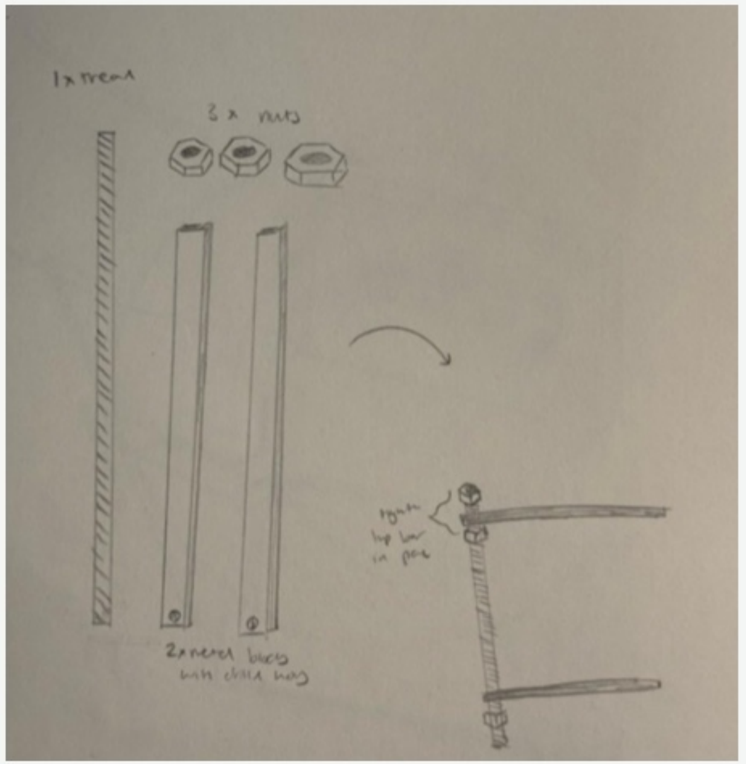
04 — Carabiner Clamp Mk2

Encloses the D-ring carabiner loop for higher tensile strength while housing a compact thumb-wheel clamping screw.



05 — Ultra-Light Mini Clip

Downscales the G-clamp to a 25mm micro-frame with a flexible split-ring tail for keyrings and cap brims.



06 — Folding Flat-Pack Clamp

Pivoting jaw arms fold flat against the screw rod to slide into a shirt pocket like a folding pocket knife.

Task 3.3 models — S-Biner clamp & belt-buckle clamp





S-Biner Clamp Prototype

1:1 cardboard model with functional spring-card gate tested clipped onto trouser belt loops for instant workshop access.

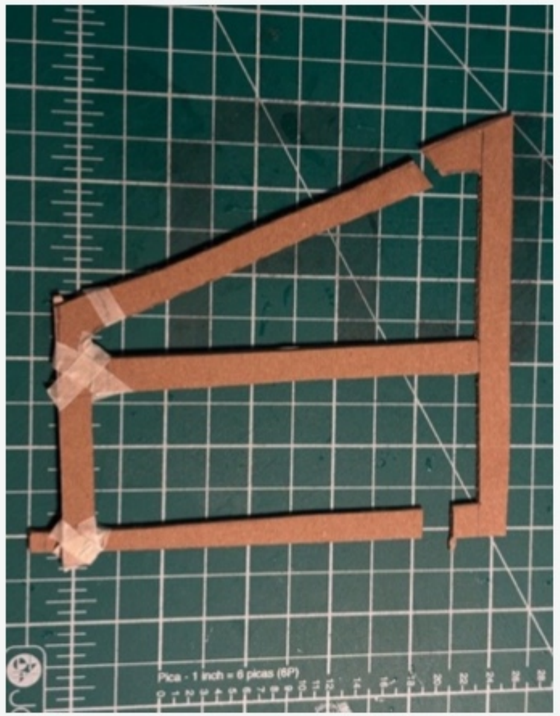


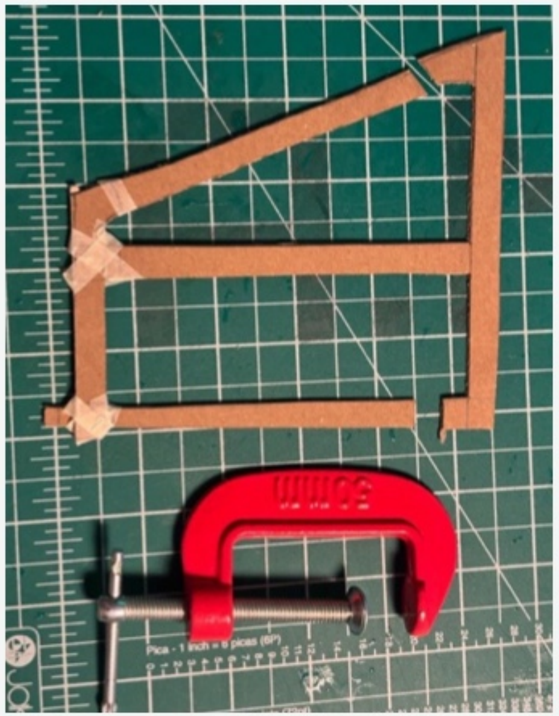


Clamp Belt Buckle Prototype

Worn threaded onto a leather belt to verify waist comfort, low profile, and rapid unbuckling when a clamp is needed.

Task 3.3 models — carabiner clamp Mk1 & Mk2





Carabiner Clamp Mk1 vs Original 50mm G-Clamp

Side-by-side comparison showing how the contoured carabiner spine reduces bulk while adding a functional spring gate.

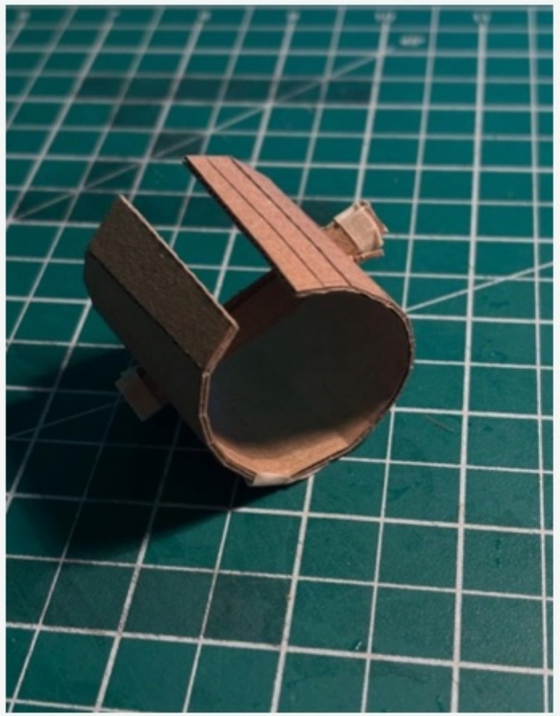




Carabiner Clamp Mk2 vs Original 50mm G-Clamp

Refined closed-loop geometry with a recessed knurled thumbwheel that eliminates the snag-prone protruding T-bar.

Task 3.3 models — mini clip Mk1 & Mk2 (everyday carry)





Mini Clip Mk1 — Flexible Ring Tail

Micro-scale clamp with an integrated flexible ring loop for keychains and zipper pulls.





Mini Clip Mk2 — Cap-Brim Wearable Utility

Ultra-compact 1:1 prototype light enough to stay clamped directly onto a baseball cap brim or jacket lapel without tipping it.